

UNIVERSIDAD
NACIONAL
DE COLOMBIA

Laboratory 5
Security

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Software Architecture

2026-II

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1. Objective

The objective of this lab is to select, analyze and implement a set of **architectural patterns** for **security**.

2. Development

Each team has been assigned one of the following security architectural patterns:

- Network Segmentation Pattern
- Reverse Proxy Pattern
- Secure Channel Pattern

Each team will work on the implementation of the assigned pattern. Using the resources previously shared with the lead architects, teams must identify a quality scenario (specifically related to security) where their assigned pattern can be effectively applied.

Once the pattern is implemented, each team will prepare a technical guide detailing the implementation process. The goal is to ensure that each pattern can later be replicated by other teams, promoting shared learning and consistent application of secure architectural practices across all projects.

3. Delivery

3.1. Deliverable

- Full name.
- Architectural view where the application of the pattern is evident.
- Technical guide, which should include:
    - A brief description of the pattern and its purpose.
    - The quality scenario addressed (e.g., confidentiality, integrity, availability).

    - The steps followed to implement the pattern in the project.
    - Any configuration or code snippets used.
    - The results or improvements observed.
    - Recommendations for other teams who may wish to replicate the implementation.

3.2. Submission Instructions

- The deliverable must be created in **.md** format and exported to **.pdf** format.
- The deliverable must be named **l5_X.pdf** (where X = the team name [1a, 1b, ..., 2a, 2b, ...]).
- The deliverable must be submitted via MiCampus UNAL.
- Only one member of the team must submit the deliverable.

3.3. Delivery Deadline

Monday, November 9, 2026, before 23h59.